LAB 7: Perform Nested and Complex Queries

Aim: Mastering Subqueries (IN, NOT IN, EXISTS, NOT EXISTS, ANY, ALL), Correlated Subqueries, Derived Tables (FROM clause), HAVING clauses, and Set Operators (UNION, INTERSECT).

1 [Subquery - IN]: Write an SQL query using an IN subquery to find the full names and countries of players who scored more than 50 runs (NRuns > 50) in any single match.

select PlayerID,FName,LName from player where PlayerID in(select PID from batting where NRuns >50);

PlayerID	FName	LName
25001	MS	Dhoni
91001	Sanath	Jayasuriya
89001	Sachin	Tendulkar

2 [Subquery - NOT IN]: Retrieve all players (PlayerID, FName, LName, Country) who have NEVER bowled in any match using a NOT IN subquery referencing the Bowling table.

select p.PlayerID,p.FName,p.LName,p.Country from player where PlayerID not in(select PID from bowling);

PlayerID	FName	LName	Country
21001	Virender	Sehwag	India
25001	MS	Dhoni	India
27001	Praveen	Kumar	India
27002	Ishant	Sharma	India
89001	Sachin	Tendulkar	India
90001	Brain	Lara	West Indies
90002	Anil	Brain	India
92001	Shane	Warne	Australia
92003	Inzamam	UI-Huq	Pakistan
93002	Heath	Streak	Zimbabwe
93003	Gary	Kirsten	South Africa
94004	Stephen	Fleming	New Zealand
95001	Ricky	Ponting	Australia
95002	Shaun	Pollock	South Africa
95003	Jacques	Kallis	South Africa
96001	Rahul	Dravid	India
96002	Herschelle	Gibbs	South Africa

	96003	Saurav	Ganguly	India
	97004	Daniel	Vettori	New Zealand
	98001	Shahid	Afridi	Pakistan
	99002	Adam	Gilchrist	Australia
	99003	Michael	Vaughan	England
	300311	Michael	NULL	NULL
	406317	Sam	NULL	India
*	NULL	NULL	NULL	NULL

3 [Subquery - EXISTS]: Write an SQL query using EXISTS to list all matches (MatchID, Ground, Date) where at least one player hit 2 or more sixes (Six >= 2).

Select o.MatchID,o.Ground,o.Date from odimatch o where exists(select MatchID from batting b where b.MatchID = m.MatchID and b.Six >= 2);

	MatchID	Ground	Date
▶	2755	Colombo	2008-08-24
	2689	Brisbane	2008-03-04
*	NULL	NULL	NULL

4 [Subquery - NOT EXISTS]: Retrieve all players (PlayerID, FName, LName) for whom NO bowling record exists in the Bowling table using a correlated NOT EXISTS clause.

select p.PlayerID, p.FName, p.LName from player p where not exists (select PID from bowling b where b.PID = p.PlayerID);

	PlayerID	FName	LName	Country
▶	21001	Virender	Sehwag	India
	25001	MS	Dhoni	India
	27001	Praveen	Kumar	India
	27002	Ishant	Sharma	India
	89001	Sachin	Tendulkar	India
	90001	Brain	Lara	West Indies
	90002	Anil	Kumble	India
	92001	Shane	Warne	Australia
	92003	Inzamam	UI-Huq	Pakistan
	93002	Heath	Streak	Zimbabwe
	93003	Gary	Kirsten	South Africa
	94004	Stephen	Fleming	New Zealand
	95001	Ricky	Ponting	Australia
	95002	Shaun	Pollock	South Africa
	95003	Jacques	Kallis	South Africa
	96001	Rahul	Dravid	India
	96002	Herschelle	Gibbs	South Africa

	96003	Saurav	Ganguly	India
	97004	Daniel	Vettori	New Zealand
	98001	Shahid	Afridi	Pakistan
	99002	Adam	Gilchrist	Australia
	99003	Michael	Vaughan	England
	300311	Michael	NULL	NULL
	406317	Sam	NULL	India
*	NULL	NULL	NULL	NULL

5 [Scalar Subquery in WHERE]: Find all players (FName, LName, Yborn) who were born before the average birth year (Yborn < AVG(Yborn)) of all registered players.

select FName, LName, Yborn from player where Yborn < (select AVG(Yborn) from player);

	FName	LName	Yborn
▶	Sachin	Tendulkar	1973
	Brain	Lara	1969
	Anil	Kumble	1970
	Sanath	Jayasuriya	1969
	Shane	Warne	1969
	Muthiah	Murlitharan	1972
	Inzamam	UI-Huq	1970
	Heath	Streak	1974
	Gary	Kirsten	1967
	Chaminda	Vaas	1974

	Stephen	Fleming	1973
	Ricky	Ponting	1974
	Shaun	Pollock	1973
	Rahul	Dravid	1973
	Herschelle	Gibbs	1974
	Saurav	Ganguly	1972
	Adam	Gilchrist	1971
	Michael	Vaughan	1974

6 [Subquery - ALL Operator]: Write a query to find player batting performances where NRuns is greater than ALL runs scored by players representing 'Australia'.

select b.MatchID, b.PID, b.NRuns from Batting b where b.NRuns > ALL (select b2.NRuns from batting b2 inner join player p on b2.PID = p.PlayerID where p.Country = 'Australia');

	MatchID	PID	NRuns
▶	2755	25001	71
	2755	91001	60
	2689	89001	91

7 [Subquery - ANY / SOME]: Find all bowling performances where Nwickets is greater than ANY wicket count achieved by bowlers from 'Sri Lanka'.

select b.MatchID, b.PID, b.Nwickets from bowling b where b.Nwickets > ANY (select b2.Nwickets from bowling b2 inner join player p on b2.PID = p.PlayerID where p.Country = 'Sri Lanka');

	MatchID	PID	Nwickets
▶	2689	99001	1
	2689	24001	1
	2755	94002	1
	2755	92002	1
	2755	23001	2
	2755	98002	3
	2689	98002	1

8 [Correlated Subquery]: Write a correlated subquery to list each player who scored the maximum runs (NRuns) among all players from their own Country.

select distinct p.PlayerID, p.FName, p.LName, p.Country, b.NRuns as MaxRuns from player p inner join Batting b on p.PlayerID = b.PID where b.NRuns = (select MAX(b2.NRuns) from Batting b2 inner join player p2 on b2.PID = p2.PlayerID where p2.Country = p.Country);

	PlayerID	FName	LName	Country	MaxRuns
▶	91001	Sanath	Jayasuriya	Sri Lanka	60
	89001	Sachin	Tendulkar	India	91
	24001	Andrew	Symonds	Australia	42

9 [Nested Subquery with HAVING]: Display players whose average runs per match is strictly greater than the overall average runs scored across all matches in the Batting table.

select p.PlayerID, p.FName, p.LName, AVG(b.NRuns) as AvgRuns from player p inner join batting b on p.PlayerID = b.PID group by p.PlayerID, p.FName, p.LName having AVG(b.NRuns) > (SELECT AVG(NRuns) FROM Batting);

	PlayerID	FName	LName	AvgRuns
▶	25001	MS	Dhoni	53.5000
	91001	Sanath	Jayasuriya	60.0000
	89001	Sachin	Tendulkar	91.0000
	24001	Andrew	Symonds	42.0000

10 [Derived Table in FROM]: Create a derived table in the FROM clause calculating total runs and total balls faced per player, then select players with a strike rate ((Total_Runs / Total_Balls) * 100) greater than 100.

select p.PlayerID, p.FName, p.LName, t.Total_Runs, t.Total_Balls, ROUND(t.Total_Runs * 100.0 / NULLIF(t.Total_Balls, 0), 2) as StrikeRate from (SELECT PID, SUM(NRuns) as Total_Runs, SUM(Nballs) as Total_Balls from batting group by PID) t inner join player p on t.PID = p.PlayerID where t.Total_Runs * 100.0 / NULLIF(t.Total_Balls, 0) > 100;

	PlayerID	FName	LName	Total_Runs	Total_Balls	StrikeRate
▶	91001	Sanath	Jayasuriya	60	52	115.38

11 [Set Operator - UNION]: Write a query using UNION to combine the PlayerIDs of players who scored > 50 runs in Batting with players who took >= 3 wickets in Bowling.

select PID as PlayerID from Batting where NRuns > 50 union select PID as PlayerID from Bowling where Nwickets >= 3;

	PlayerID
▶	25001
	91001
	89001
	98002

12 [Set Operator / Subquery Intersect]: Find players who both batted AND bowled in MatchID = 2755 using an INTERSECT query or equivalent subquery structure.

select PlayerID, FName, LName from player where PlayerID in (select PID from batting where MatchID = 2755) and PlayerID in (select PID from Bowling where MatchID = 2755);

	PlayerID	FName	LName
▶	94002	Chaminda	Vaas
	92002	Muthiah	Murlitharan
	91001	Sanath	Jayasuriya
	23001	Yuvraj	Singh
	98002	Harbhajan	Singh

13 [Complex Grouping & HAVING]: Display Ground, total runs scored on that ground, and total matches played, considering only grounds where more than 2 matches took place and total runs exceed 100.

select m.Ground, COALESCE(SUM(b.NRuns), 0) as TotalRuns, COUNT(DISTINCT m.MatchID) as TotalMatches from odimatch m left join batting b on m.MatchID = b.MatchID group by m.Ground having COUNT(DISTINCT m.MatchID) > 2 and COALESCE(SUM(b.NRuns), 0) > 100;

	Ground	TotalRuns	TotalMatches
▶	Colombo	153	3